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Wisdom/Madness of Crowds and Perils of Point Forecasts

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Abstract. This is not primarily a methods paper, but rather a paper about the limitations of methods. Using a large data set of expert forecasts with realizations, it emerges that expert point forecasts vary widely, and the distribution of point forecast errors appears to be very fat tailed. This paper uses simulated data to show that when samples from fat-tailed distributions are averaged, the results may not converge: extreme "outliers" arise with a frequency sufficient to disrupt any trends toward (apparent) convergence. It also shows that real-world expert judgment data from a number of different domains, collected over a period of decades, exhibit such fat-tailed behavior as regards point forecasts. Methods based on the "wisdom of crowds" that aggregate point forecasts of multiple experts are unlikely to yield good performance in such cases, regardless of the precise method by which judgments are weighted and aggregated. Moreover, increasing the number of experts used in such studies does not improve performance but rather degrades it, as does increasing the "diversity" of expert panels. However, when experts provide probabilistic forecasts, both the number of experts and diversity are helpful for the panel's overall statistical accuracy, which in turn correlates positively with lower forecast error. The moral of the story is that point estimates elicited from multiple experts should not be aggregated. Decision analysts should instead rely on aggregating probability distributions elicited from experts rather than point estimates, as is already considered best practice. Weighting based on the experts' measured performance yields even better results.

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Keywords: wisdom of crowds • mean absolute percentage error • statistical accuracy • expert judgment • fat-tailed distributions • dependent expert errors

1. Introduction

The notion that crowds possess wisdom dates back to antiquity but was challenged with Mackay's (1841) *Memoirs of Extraordinary Popular Delusions and the Madness of Crowds*. Galton (1907) parried with Cornwall fair goers' average (originally median) estimate of the dressed weight of a bull, which was nearly spot on. His interest was not limited to bulls: "In these democratic days, any investigation into the trustworthiness and peculiarities of popular judgments is of interest (p. 450)." In these days of artificial intelligence and machine learning, interest in "trustworthiness of popular judgments" quickens. In 1907, there was discussion whether Galton's (1907) "vox populi" should not be called "vox expertorum" given the large number of expert butchers and farmers attending these events. Surowieki's (2005) *New York Times* best seller, *The Wisdom of Crowds*, distinguished wise crowds from irrational crowds on five criteria: diversity, independence, decentralization, aggregation, and trust. Murray (2019) replied with *The Madness of Crowds: Gender, Race and Identity*. Much is written on the credibility of crowds, but the discussions are largely anecdotal. For this reason, the 565,000 plus hits for "wisdom of crowds" on Google Scholar are not reviewed here. There has been valuable psychometric research focusing on point versus interval versus density

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forecasts (see, e.g., Bruine de Bruin et al. 2011, Niu and Harvey 2022) highlighting concerns similar to those driving the present research. These studies employ psychometric experiments rather than real expert field data, use scale-dependent error measures, and in some cases assume that an interval forecast is the average of the interval's endpoints—features that prevent translation to expert field data. Nonetheless, the common interest in forecast errors is noteworthy. Equally noteworthy is interest in defining and measuring expertise, as evinced in Thomas and Lawrence (2018), Weiss and Shanteau (2004), Hanea et al. (2018), and Burgman et al. (2011).

Expert probabilistic forecast data for which realizations or true values are available at scale are valuable for several reasons: (1) the inevitable winnowing of reliable crowds often turns on properties associated with "vox expertorum"; (2) probabilistic forecasting converts all quantities to a common scale, namely, probability, because of which (3) we can develop scale independent performance metrics; and finally, (4) we have extensive data from structured expert judgment (SEJ) panels. To date, 107 SEJ studies have been performed with traceable publications.[1] All these studies have no common experts and may be considered independent. A few large studies involving repeated elicitations with overlapping expert panels (Aspinall et al. 2002, Aspinall 2006, Hald et al. 2016) are excluded. The 107 panels consist, on average, of 11 vetted experts giving 5, 50, and 95 percentiles for uncertain variables from their fields and also for, on average, 14 calibration variables from their fields to which true values are or become known. Performance on these calibration variables is used to construct performance-weighted combinations of expert cumulative distribution functions (CDFs) and compared with equally weighted combinations. Expert performance is persistent (Cooke et al. 2021), and performance-based combinations are superior to equal-weight combinations both in (Colson and Cooke 2018) and out of sample (Eggstaff et al. 2014, Colson and Cooke 2017) and have been evaluated in real applications (Aspinall 2010). The expert data are publicly available at https://rogermcooke.net/.

Among the alternative approaches, Lichtendahl et al. (2013) (see also Winkler et al. 2019) advocate quantile averaging. Quantile averaging has the advantage that point forecasts can be obtained from a set of experts simply by averaging the experts' medians (50% quantiles). It is not necessary to assess each expert's CDF and form a combined CDF to extract a point forecast. Averaging expert medians is very different from averaging experts' CDFs and extracting the median from the average CDF, as will appear. Cooke (2022) showed that quantile averaging is equivalent to taking the harmonic mean of densities; it shrinks variance relative to an equal weight average of CDFs and significantly degrades statistical accuracy (SA). Of the 49 panels in that study, the hypothesis that the quantile average combinations were statistically accurate would be rejected at the 5% level in 28 studies. For the equal-weight and performance-weight combinations, two and three studies, respectively, would be rejected at the 5% level. The informativeness of the quantile averages and performance-weighted averages of CDFs were comparable. The mean absolute percentage error (MAPE) of the average of experts' medians was much greater than that of the median of equally weighted combinations of experts' CDFs. Better still were performance-weighted combinations (Cooke et al. 2021; see also Section 6).

Crowd-casting offers a perspective on expert forecasting panels: Scientists in an SEJ forecasting panel have similar training, follow the same literature, and often know each other. Surowieki (2005, p. 36) states: "Homogeneous groups, particularly small ones, are often victims of what the psychologist Irving Janis called 'groupthink.'" "After a survey of expert forecasts and analyses in a wide variety of fields, Wharton professor J. Scott Armstrong wrote, 'I could find no studies that showed an important advantage for expertise'" (Surowieki 2005, p. 33). The antidote is crowd size: "much of what we've seen so far suggests that a large group of diverse individuals will come up with better and more robust forecasts and make more intelligent decisions than even the most skilled 'decision maker'" (Surowieki 2005, p. 32). "Crowds" can include diverse collectives beyond expert forecasting panels, but expert panels certainly fall within the wisdom of crowds (WOC) discussion. As such, it is relevant to ask whether research in expert forecasting panels can shed light on the putative wisdom of crowds.

This study uses SEJ data to examine two pillars of WOC: (1) Is crowd size beneficial? (2) Is "diversity"/"independence" beneficial? To address these, 40 forecasting panels with at least 10 experts and at least 10 calibration variables are selected, giving 586 forecast

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variables with realizations, 698 experts, and 10,189 expert forecasts (see Appendix A, Table A.1). The complementary set (67 unselected studies[2]) with fewer than 10 experts or fewer than 10 calibration variables per panel involve 842 forecast variables with realizations. Studying performance in these two data sets allows us to study the effect of panel size. "Beneficial" is measured by two performance metrics, MAPE and SA. The conclusion drawn from this study of expert panels is that extracting point forecasts from a combination of expert probability distributions is better than combining expert point forecasts. This entails that crowd-casters might better ask the crowd members to quantify their uncertainties, and combine these, rather than asking for point forecasts.

Section 2 introduces the performance metrics, Section 3 gives an intuitive introduction to fat-tailed distributions, Section 4 discusses the effect of crowd size in our data cohort, and Section 5 discusses independence and diversity and defines *net agreement* for point forecasts and for uncertainty bands. Section 6 considers the influence of covariates on the performance metrics, and Section 7 concludes. Appendix A tabulates the panels in this data cohort, proves that WOC is slightly better than picking a random expert, provides mathematical definitions of net agreement, and gives an illustrative calculation. Appendix B is a short primer on fat tails focusing on important definitions and properties without proofs. Fat tails are studied in a branch of mathematics called extreme value theory. They are not generally covered in undergraduate statistics or economics programs and are often counterintuitive: Historical averages may lack any predictive value, and sample means, correlations, and covariances may not converge. Many of these counterintuitive features emerge in "black swan" discussions (see Taleb 2007, Cooke et al. 2014). The main text seeks to cultivate intuitions while avoiding formal mathematics; Appendix B is recommended for those wishing to go deeper into the subject.

2. Performance Metrics

There are many measures for proximity of point forecasts to a realization (Gneiting et al. 2007, Morley et al. 2018), each with pros and cons. The absolute percentage error (APE) and its mean (MAPE) are perhaps the most popular. For forecast f with realization r, the absolute percentage error is $|(f-r)/r| = |f/r-1|$ and is unstable for r close to zero. It is scale invariant, so scores for different forecasts and different realizations can be averaged yielding the MAPE. We can average over all calibration variables for each expert to form an expert MAPE, and we can average the expert MAPEs for all experts in a panel to arrive at an expert panel MAPE. This is the expected MAPE of an expert randomly chosen from the panel for a randomly chosen variable. Invoking the WOC, we can first average experts' "best guess" forecasts and then compute the WOC MAPE, per variable and per panel. Absent an alternative, the median is considered the best guess. By Jensen's inequality, the WOC panel MAPE is always less or equal to the expert panel MAPE, though the mean difference per panel is a mere 0.009 in this case (see Appendix A, Table A.1). Morley et al. (2018) note that "Underprediction is … less heavily penalized than overprediction (p. 73)." Indeed, if the quantity is nonnegative, then the most severe underprediction arises when $f=0$, in which case the APE = 1. For overprediction, the APE is unbounded.

Other dimensionless measures have been proposed. Flores' (1986) proposal $2|(f-r)/(f+r)|$ has, according to Morley et al. (2018), an "unintuitive normalization (p. 71)." Indeed, if $f>r>0$, then this score is bounded by two. Somewhat better is the symmetrized MAPE, denoted by *Sym MAPE*:

$$(f-r)^2/(f\times r) = r/f - 1 + f/r - 1,$$

whose tail fatness is compared with that of MAPE in Appendix B.

From the viewpoint of statistical decision theory, point forecasts should be extracted from the probability distribution of the unknown quantity so as to minimize expected loss. This is not the viewpoint from which proximity measures like MAPE emerge. Indeed, no distribution for r is available, and each expert has his or her own subjective probabilities. If $r \neq 0$, the optimal value of MAPE is zero. We should hope that the average MAPE over a large number of expert forecasts of nonzero variables should tend toward the optimal value zero. Any other limiting value would signal expert bias. In fact, as we shall see presently, they appear to be neither biased nor unbiased, but dissipative.

SA is based on the relative frequency with which the realizations of independent calibration variables

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fall inside the forecaster's four interquantile intervals. SA is the probability that these relative frequencies should differ from the theoretical interquantile probabilities (5%, 45%, 45%, 5%) by at least the observed amount on the hypothesis that the expert is statistically accurate and interquantile hits are independent.[3] Low values near zero mean that it is very unlikely that the forecaster's probabilities are statistically accurate; high values, near one, indicate good agreement between observed and expected relative frequencies. This is a standard chi-squared test for goodness of fit; detailed recent explanations are available in (Cooke et al. 2021, especially the supplementary information).

3. Tail Size[4]

WOC discussions such as those cited above implicitly invoke fragile statistical assumptions. If we sample a finite set of numbers from some distribution, we can always compute the average of these numbers as well as the variance, standard deviation, correlations with other sets of numbers, etc. However, if we sample more numbers or sample a like-sized second batch, do these averages, variances and correlations tend to agree? The law of large numbers says that averages, variances, and correlations stabilize as we draw ever larger samples; however, this law applies only if the distribution from which the numbers are drawn has finite variance. If the distribution has infinite variance, then the law of large numbers does not hold (see Appendix B).

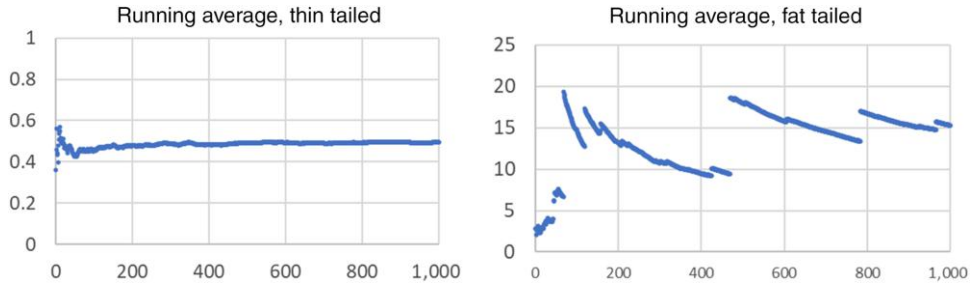
Pictures illustrate the difference between fat- and thin-tailed distributions. The left panel of Figure 1 gives running averages (average the first two, then the first three, etc.) of 1,000 independent samples from a uniform distribution on the [0, 1] interval. The horizontal axis gives the size over which the average is taken. On the vertical we plot the running average. At the horizontal value 1,000, we average all 1,000 samples. In the right panel, we do the same, with the same numbers, except that these numbers are now inverted; 0.1 becomes 10, etc. The inverse of the uniform distribution is a very fat-tailed distribution. It has a Pareto(1) distribution on [1, ∞) with tail index one for which $\text{Prob}(X > x) = x^{-1}$. Its mean is infinite. Pareto distributions are tail limits of regularly varying distributions, belonging to the class of subexponential distributions for which the sum and maximum are asymptotically equivalent (see Appendix B).

With thin-tailed distributions, running averages converge; with fat-tailed distributions like Pareto(1), they do not. Very large values keep popping up at a rate that prevents convergence. Fat-tailed distributions are not uncommon, but are not common knowledge.

If we draw repeated samples of size 1,000, the thin-tailed running averages will differ a bit at the beginning but quickly settle into the pattern. Figure 2 shows what happens with three samples of 1,000 from the distributions in Figure 1. Notice the changing scale on the vertical axis for the fat-tailed distributions; these samples do not settle into a pattern. The are dissipative. Notice that if you only look at the middle graph in the lower row of Figure 2, you might think that the data are not stationary and that the data reflect a temporal trend. However, these data are sampled independently from the same distribution.

Statisticians do not like fat tails as they prevent application of the familiar statistical methods. It is tempting to delete a few large values as "outliers" so that the rest "look normal." The U.S. National Flood Insurance Program actuaries who set insurance rates gave 1% weight to losses from Hurricanes Katrina, Wilma, and Rita "to reflect the events of 2005 without allowing them to overwhelm the pre-Katrina

Figure 1. (Color online) (Left) Thin-Tailed and (Right) Fat-Tailed Running Averages



Note. Horizonal axes denote the number of samples over which we average.

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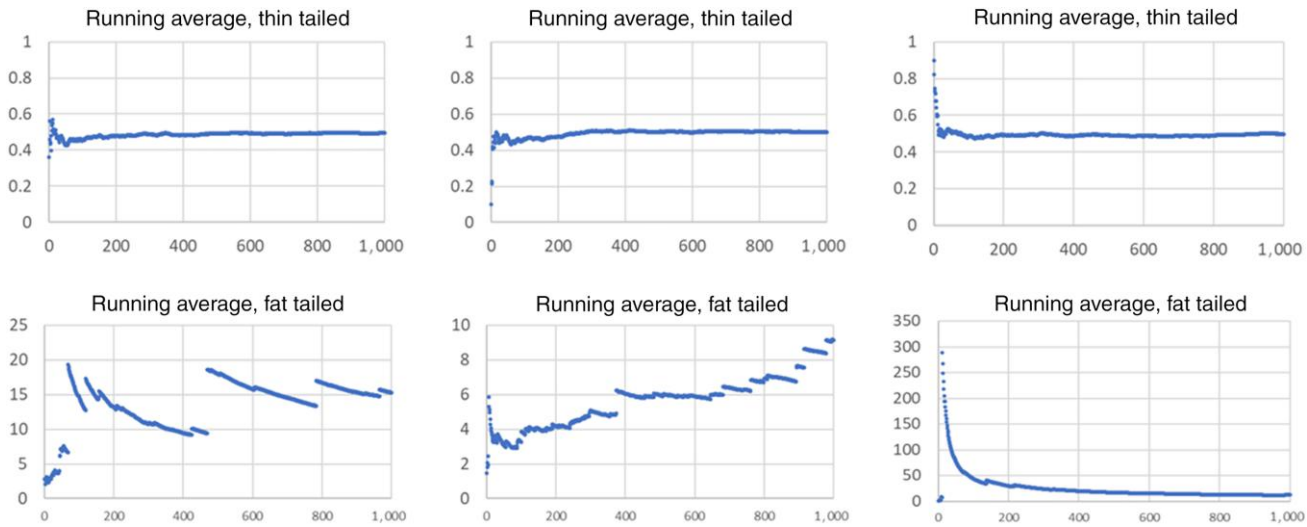
Figure 2. (Color online) Three Repeated Samples with Running Averages

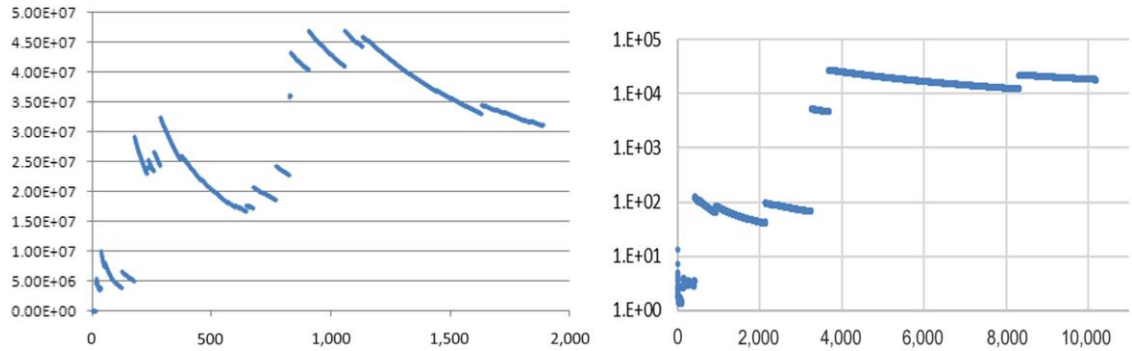


experience of the Program" (see Cooke et al. 2014, p. 11). However, when one looks at more samples from a fat-tailed distribution, one realizes that the large values are characteristic of the whole distribution. If we order the sample from smallest to largest values, we see that the distance between adjacent samples tends to get much larger as the sample values get larger. As we gather more samples, the average of the whole sample is driven by the largest sample in the set. In fact, this is a defining feature of the "subexponential" class of fat-tailed distributions. Once we look, we can see fat-tailed distributions everywhere—damages from natural disasters, crop insurance claims, citation scores, flood damages, income distributions, hospital discharge rates. Many loss distributions in risk analysis fall into this category (Cooke et al. 2014). What about experts' errors?

4. Crowd Size

Do averages of ever more absolute percentage forecast errors trend down to the optimal value zero? There is an antecedent question: Do such averages converge at all? Figure 3 shows running averages of U.S. damages in excess of $10 million due to natural disasters (left) and absolute percentage error in 10,198 expert forecasts (right). To be sure, these experts assess different quantities, but their absolute percentage errors can be pooled and plotted on an absolute scale. This reflects the factor by which the forecast differs from the realization in absolute value for an expert forecast drawn randomly from this pool. Such graphs depend on the ordering, but random reorderings will exhibit the same key feature: ever larger values keep popping up that prevent convergence. Note the logarithmic scale on the right panel.

Figure 3. (Color online) (Left) Running Averages for U.S. Damages in Excess of $10 Million (Cooke et al. 2014) and (Right) Running Averages for 10,189 Expert Absolute Percentage Errors



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Experts' absolute percentage errors in aggregate appear very fat tailed, and averages do not appear to converge. Does this also apply to WOC forecasts? Figure 4 (top) shows 586 realizations in ascending order plotted with their WOC forecasts. Note the very small realization with forecast differing by eight orders of magnitude. To avoid instabilities due to small realizations, we subset the 535 forecasts for which the realizations are greater or equal to 0.1. The running averages are shown in Figure 4 (bottom); the averages of ever larger sets of absolute percentage forecast errors just keep growing.

The WOC panel MAPEs in Table A.1 of Appendix A illustrate the "sum resembles the max" property of fat-tailed distributions. The sum of all 40 WOC panel MAPEs is 928,834.49; the largest WOC panel MAPE is from the Portuguese volcano study SETE_CIDADES, 877,202.18. The second largest is from the Japanese volcano study TOPAZ, 49,777.81. If we drop these two studies, the sum drops to 1,854.51. Perhaps the whole fat-tail problem lies with volcanology? Ranking the 40 studies for WOC panel MAPE, 13 of the 40 are volcanology studies whose mean rank is 23.46 (a bit above average for this cohort) and mean number of experts is 15.9 (a bit below cohort average). The Mann–Whitney test (normal approximation) cannot distinguish the volcanology studies from the others based on the ranking of their MAPEs.

Figure 4. (Color online) (Top) Five Hundred and Eighty-Six WOC Forecasts vs. Realizations and (Bottom) Running Averages of 535 WOC MAPEs, RLS Denotes Realization

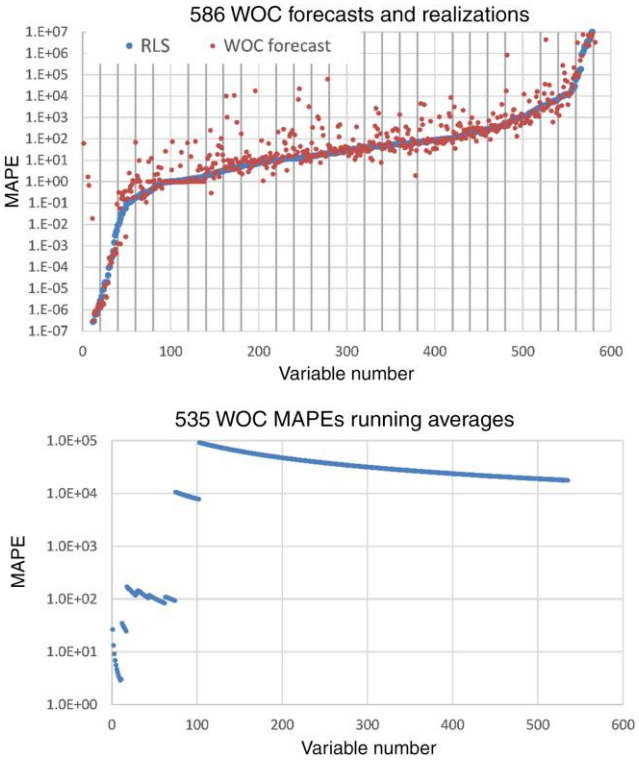

Figure 5 plots all 586 WOC MAPEs against the panel size over which the median forecasts are averaged. The rank correlation in Figure 5 is weakly increasing in panel size. The obesity index, defined in Appendix B, offers a quantitative line of attack, which has proven useful in environmental science (Wietzke et al. 2020, Gupta and Chavan 2021). This index reflects the degree to which the max resembles sum, both for finite samples and for full distribution functions, not just for behavior as the underlying variable x goes to infinity. It may be conceived as measuring a "degree of subexponentiality" analogous to the tail index for regularly varying variables. The obesity of a finite sample is estimated by bootstrapping based on random 4-tuples. The number of 4-tuples grows quickly (there are 4.86E9 4-tuples of the 586 MAPEs), enabling estimates with statistical power. The obesity of an empirical sample may be compared with the obesity of familiar fat- and thin-tailed distributions, like the Pareto(α) with tail index α or the exponential. With 10,000 sampled 4-tuples, it emerges that the obesity of the 586 MAPEs is significantly greater than that of a Pareto(1). The Sym MAPE obesity is significantly lower than that of the standard MAPE but still significantly greater than Pareto(1). Obesity of the 842 MAPEs from the complementary set of 67 studies ($107 - 40$) is also significantly lower than the obesity of the 586 MAPEs, though still greater than that of a Pareto(1). With regard to Figure 5, distinguishing the obesity of MAPEs of those panels with 10–19 experts from that of those with at least 20 experts requires 50,000 samples (see Appendix B). In sum, there is ample statistically significant evidence that panel size increases obesity with regard to MAPE, though even smaller panels still have a fat-tail problem. Needless to say, it is impossible to extrapolate beyond the current pool of expert forecasts.

5. Dependence and Diversity

Increasing crowd size can have no effect if forecasters all say the same thing. Diversity and dependence must

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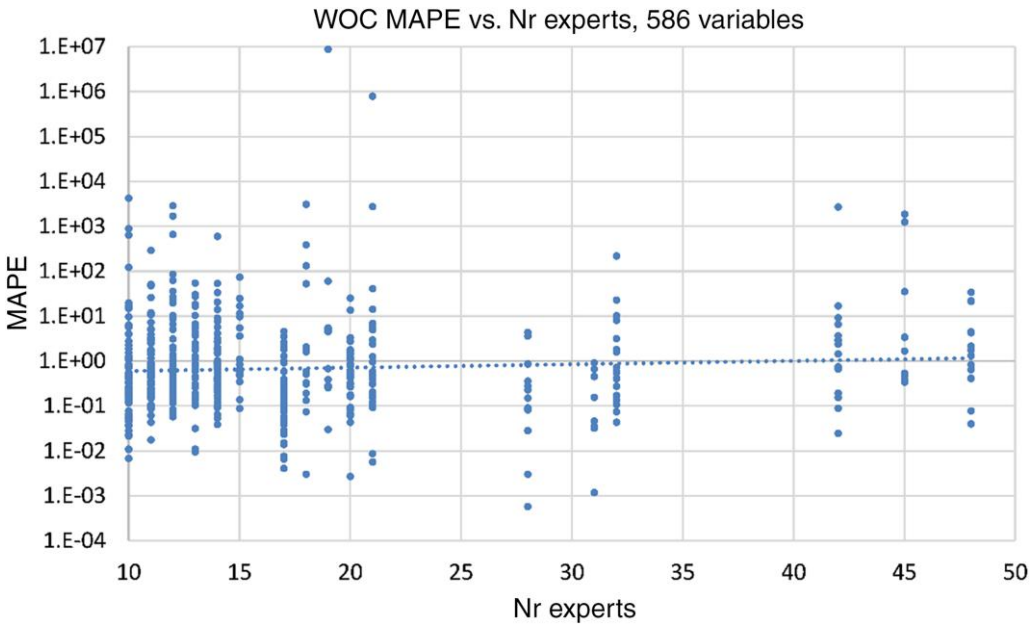
Figure 5. (Color online) WOC MAPE Against Number of Empaneled Experts for 586 Variables



be addressed. If we look only at expert forecasts in a panel, then of course they will be dependent for the simple reason that they are forecasting the same quantity. All forecasts of a volcanic eruption in cubic meters will be large; all forecasts of the weekly growth of the dome in meters will be small. Apparently, that is not the right question. We should be asking about dependence in experts' forecast *errors* relative to the realizations. Absent realizations, discussions about "dependence and diversity" are untethered and refer to things like "people's opinions are not determined by the opinions of those around them (p. 10)" (Surowieki 2005). With realizations, diversity can be operationalized as the amount of disagreement in a panel.

To capture diversity and dependence, four dependence matrices are constructed for each panel. The *density@realization* (*dens@rls*) matrix assigns each (row = expert, column = variable) the (interpolated) percentile of the expert's probability distribution realized by the true value. The matrix for CO2em (Rennert et al. 2022) is shown in Table 1 with 10 experts and 11 calibration variables. We can compute correlations variable-wise or expert-wise. It emerges that the mean correlation expert-wise is 0.65, and variable-wise it is –0.069 (see Appendix A, Table A.1).

The *HiLo* diversity matrix (not shown) assigns the value –1 to an (expert, variable) if the realization is below expert's point forecast (median) and assigns 1 otherwise. Total agreement (minimal diversity) entails that expert medians are either all above or all below the realizations. The *Tail* diversity matrix assigns –1 if the realization falls outside the expert's 90% confidence band and 1 otherwise. Total agreement (minimal diversity) means that all experts' confidence bands catch or all fail to catch the realizations. For two experts, the net Tail or HiLo agreement (agreements minus disagreements) is just the inner product of the two experts' rows. The net agreement for {–1, 1} matrices can be computed from the covariance matrices (see Appendix A with illustrative calculation in Table A.2) and give an intuitive rendering of the different aspects of dependence. To compare across studies, the net agreement is divided by the numbers of experts and variables. The MAPE matrix assigns the absolute percentage error to each (expert, variable).

A simple analysis of variance decomposes the overall variance V in each matrix into components contributed by experts e and by variables v, as illustrated for Table 1. The term $E(e|v)$ denotes conditional expectation of e given v; these are the column averages given in the last row of Table 1. The term $V(E(e|v))$ denotes

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Table 1. Dens@realization for CO2em

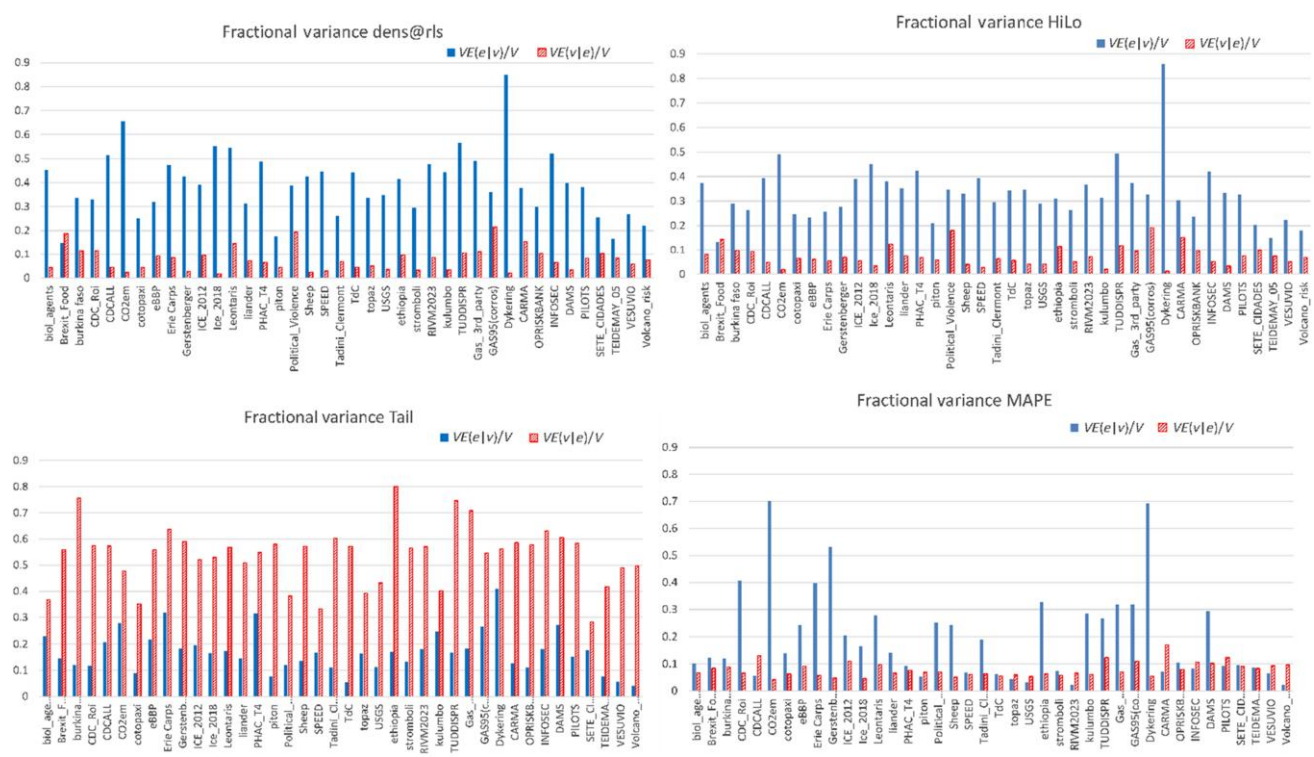
Percentile RLS	vbl1	vbl2	vbl3	vbl4	vbl5	vbl6	vbl7	vbl8	vbl9	vbl10	vbl11	$E(v\|e)$
exp1	0.32	0.536	0.02339	0.5864	0.04445	0.3875	0.95	0.97833	0.04975	0.52775	0.15485	0.4144
exp2	0.725	0.04899	0.02865	0.04779	0.04445	0.62273	0.96576	0.98787	0.04975	0.95485	0.42463	0.4455
exp3	0.9568	0.455	0.02865	0.03955	0.03079	0.32	0.95778	0.98787	0.04975	0.96078	0.04742	0.43949
exp4	0.95822	0.26	0.0204	0.7955	0.04445	0.97585	0.97816	0.94254	0.01866	0.96078	0.2597	0.56493
exp5	0.47	0.0445	0.01711	0.04779	0.78125	0.275	0.6125	0.98189	0.49454	0.9612	0.95034	0.51237
exp6	0.58077	0.11	0.02865	0.4682	0.575	0.39412	0.44706	0.98787	0.02713	0.59226	0.56041	0.43377
exp7	0.66364	0.34	0.104	0.57092	0.03872	0.04561	0.17857	0.98444	0.02713	0.95245	0.10243	0.36436
exp8	0.455	0.365	0.02865	0.4523	0.0875	0.725	0.77	0.98444	0.27295	0.95368	0.15485	0.47722
exp9	0.89	0.32	0.04322	0.64184	0.125	0.5	0.77692	0.98444	0.02713	0.58325	0.5699	0.49652
exp10	0.77	0.266	0.095	0.4523	0.60761	0.275	0.5	0.98907	0.19727	0.67775	0.23836	0.46894
$E(e\|v)$	0.67894	0.27455	0.04177	0.41026	0.23792	0.45208	0.71367	0.98088	0.12141	0.81248	0.35529	0.46175

Note. Exp, Expert; vbl, variable; RLS, realization.

the variance in these column averages. Similarly, $E(v \mid e)$ denotes the row averages in the last column of Table 1; the variance of the row averages is $V(E(v|e)$. The fraction of the total variance V explained by variation over variables, $V(E(e \mid v))/V$, is much greater than that explained by experts, $V(E(v \mid e)/V$. This is because the experts are much more correlated than the variables, 0.65 versus –0.069.

The fractions of explained variance for all studies and all four dependence matrices are shown in Figure 6.

Figure 6. (Color online) Fractions of V Explained by Variation over Variables ($V(E(e|v)/V$) and by Variation over Experts ($V(E(v|e))/V$)



Notes. The variance decomposition for the matrices dens@rls, HiLo, and MAPE are similar and indicate moderate clustering of experts. For Tail, the pattern is reversed; more variance is explained by experts, indicating greater independence of experts with regard to the 90% intervals. MAPE is similar to dens@rls and HiLo, but explains less of the total variance (https://www.cooke-aspinall.net/).

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A remarkable feature emerges: The variance decompositions for the matrices dens@rls, HiLo, and MAPE all show that variables explain more variance than experts, although variables and experts explain less of the total variance in the MAPE matrix. These three matrices show a moderate clustering of experts, which explains the experts' lower fraction of explained total variance. For Tail, the pattern is reversed; more variance is explained by experts. The experts' placement of medians is relatively dependent, whereas the experts' ability to catch the realizations in their 90% bands is much less so. If we are looking for independence/diversity, we should be looking at the Tail matrix.

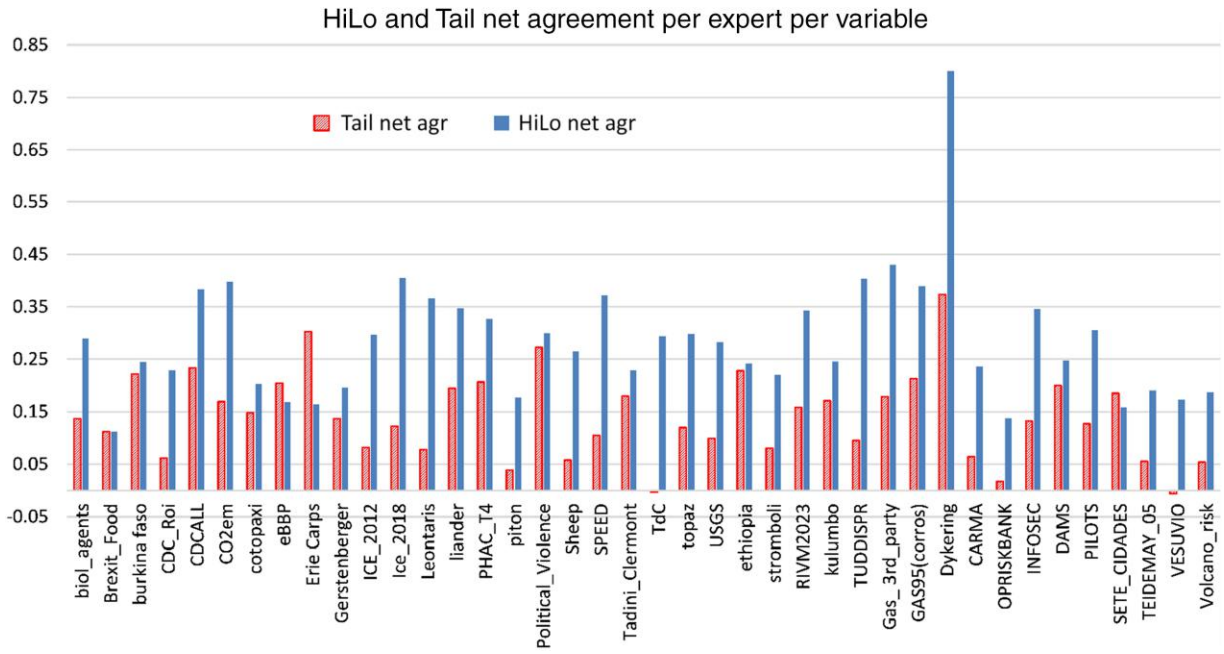
Figure 7 reinforces this fact. The net agreement per expert per variable is notably greater for the HiLo than for the Tail dependence matrix. Note that the net agreement can be negative, as we see for Tail in the VESUVIO study. Figure 8 compares the net agreement in these two cases with the expected net agreement if the 1s and –1s in these matrices were assigned to cells randomly (see Appendix A). For HiLo, the net agreement is much greater than we should expect if the agreements were at random, whereas the Tail net agreement is closer to the expected random agreement, though still greater.

6. Influence of Crowd Size and Diversity on Performance Metrics

Section 4 showed that increasing panel size increases tail obesity, exacerbating nonconvergence of WOC panel MAPE. As a result, larger panels tend to have larger mean absolute percentage errors. Traditional tools like regression and product moment (Pearson) correlation invoke mean values and covariances, which may tend toward infinity. Rank (Spearman) correlation converts the original scales to orderings on the [0, 1] interval and thus avoids convergence issues. Table 2 gives rank correlations of the two performance metrics with study covariates. Statistical significance is based on the Student T approximation to the distribution of rank correlation (Kendall et al. 1939).

Significance level 0.05 is "significant," significance levels in (0.05, 0.2] are "indicative," and others are weak. Statistical significance is challenging because of the limited number of studies (40) and the large

Figure 7. (Color online) Net Agreement per Expert per Variable for HiLo and Tail Dependence Matrices



Note. agr., Agreement.

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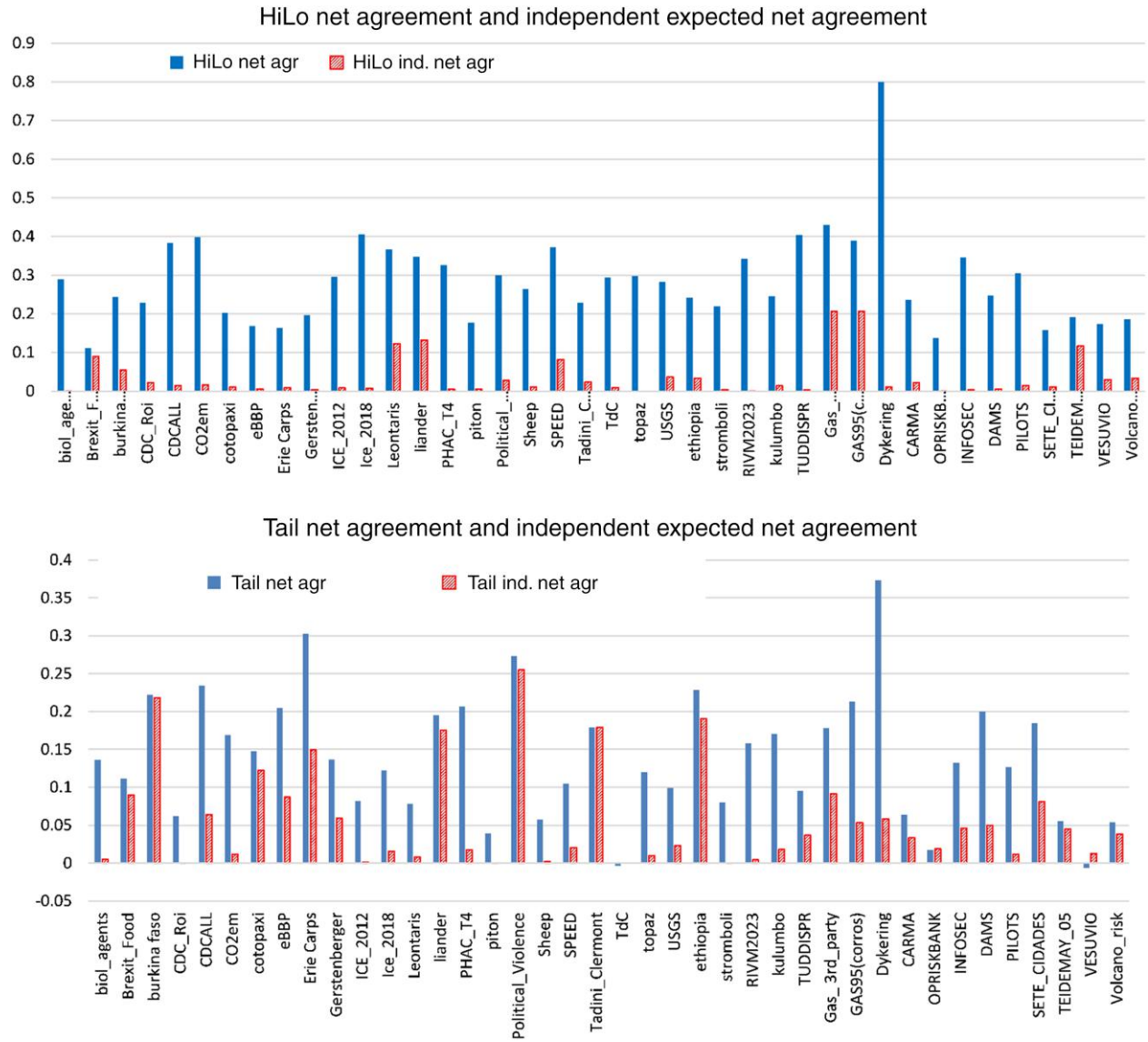
Figure 8. (Color online) Comparison of Net Agreement and Expected Net Agreement if the Agreements Were Random, for HiLo and Tail Dependence Matrices



Note. agr., Agreement; ind., independent.

number of ties in number of experts (nrex) and number of variables (nrvb; see Table A.1). Only the negative correlation between SA and HiLo net agreement is significant at the 5% level. Greater HiLo net agreement corresponds to lower (worse) SA. Translating *disagreement* as "diversity," diversity is associated with high SA. Because SA is also negatively correlated with MAPE, one might think that HiLo net agreement would be positively correlated with MAPE ("the enemy of my enemy is my friend"). However, the correlation of MAPE with HiLo net agreement is also negative, though not significant. This suggests that SA may be partially masking the effect of HiLo net agreement on MAPE. These issues can be overcome by reverting to the full cohort of 107 studies. Table 3 shows that the positive rank correlations with nrex are significant at the 0.01 level in this larger sample. More experts is associated with higher (worse) MAPE. Note

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Table 2. Rank Correlations of Performance Metrics WOC Panel MAPE and Mean SA, with Significance Levels

	40 studies; rank correlation with:				
	Mean SA	Tail net agreement	HiLo net agreement	nrex	nrvb
WOC panel MAPE	−0.20	−0.05	−0.15	0.19	0.04
Significant at	0.11	>0.20	0.18	0.12	>0.20
Mean SA		−0.07	−0.26	0.17	0.09
Significant at		>0.20	0.05	0.15	>0.20

that nrvb is not significantly correlated with either performance metric. Note also that the product moment correlations are all negligible. When variances go to infinity, product moment correlations typically go to zero.

Section 5 noted that Tail net agreement is weaker than HiLo net agreement and closer to independent tail agreement. The different impacts of these two types of agreement can be seen by averaging their values over deciles of the WOC MAPE distribution for the 40 studies. Figure 9 shows that HiLo agreement is greatest for the lowest (best) 10% of the MAPE values and decreases as WOC MAPE values get larger (worse). More agreement corresponds to lower (better) MAPE values. Tail agreement shows a much weaker association.

Focusing on the empaneled expert with the best SA yields a sharp relationship. As shown in Figure 10, the Spearman rank correlation between the SA of the expert with the best SA in each panel and this expert's MAPE is –0.45 and is significant at the 0.002 level.

A previous study (Cooke et al. 2021) used 49 panels with 570 calibration variables and 530 experts.[5] This is a different subset of the 107 studies, containing 28 panels with fewer than 10 experts, and for this reason provides a useful comparison. Averaged over all 570 calibration variables, the MAPEs were 2.2 for performance-weighted combinations, 3.8 for equal-weighted combinations, and 1472.3 for WOC (average of medians). In the 40 panels of the present study, the average WOC panel MAPE was 23,220.86, underscoring the *disadvantage* of larger panels for WOC (Appendix A, Table A.1). Instead of averaging the experts' medians per panel, if we chose the expert with the best SA, the MAPE over the 40 panels would be 6.3. Figure 11 captures the relationships in a simple dependence diagram. High HiLo diversity (low HiLo agreement) is detrimental for obtaining low WOC panel MAPEs, and helpful for obtaining high statistical accuracy. A high number of experts is detrimental for obtaining a low WOC panel MAPE and helpful for obtaining high statistical accuracy. All the while, high statistical accuracy is helpful for achieving a low WOC panel MAPE.

7. Conclusions

Physicist Max Planck (1950) famously quipped "science advances one funeral at a time (p. 33)," reenforcing the "group think" risk of small expert panels. Au contraire, says Oreskes (2019) in *Why Trust Science?*: scientific consensus resulting from rigorous peer review provides a basis for trust. Discussions about crowd wisdom and/or expert wisdom can move beyond anecdotes by analyzing data of forecasts with realizations. For expert

Table 3. Rank and Product Moment Correlations in the Full Cohort of 107 Studies

	Expert MAPE	WOC MAPE	Significant at
Rank correlation with nrex			
Product moment correlation	−0.07	−0.07	
Rank correlation	0.24	0.23	0.01
Rank correlation with nrvb			
Product moment correlation	−0.01	−0.01	
Rank correlation	0.02	0.04	

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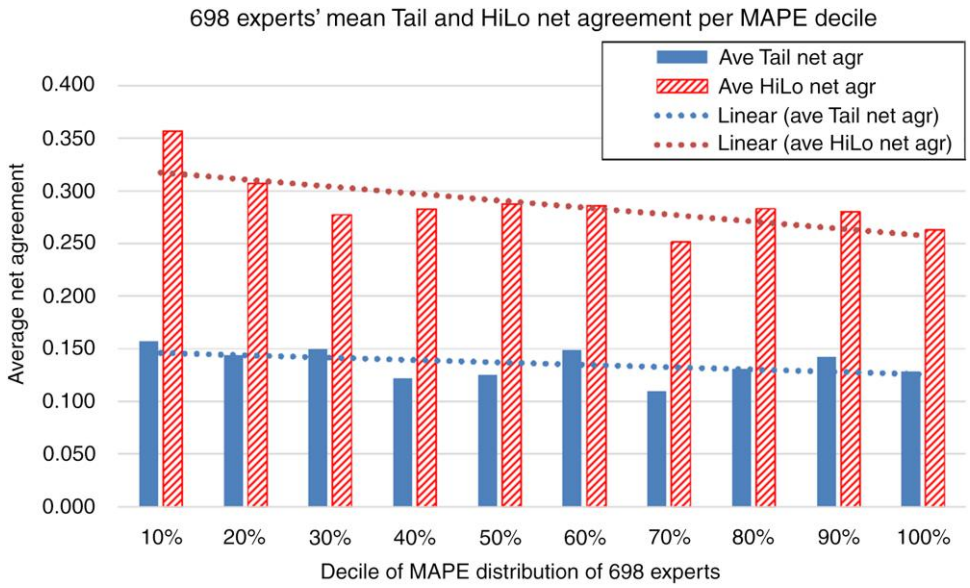
Figure 9. (Color online) Expert Tail and HiLo Agreement Scores Averaged over Each Decile of the Expert MAPE Distribution



panels, such data are publicly available at https://rogermcooke.net/. Publications of SEJ applications are available at https://www.cooke-aspinall.net/. Both expert MAPEs and WOC MAPEs appear to be very fat tailed, raising doubts about WOC MAPE convergence. WOC MAPE is slightly better than picking a random expert. Crowd size and HiLo diversity work in opposite directions on the two performance metrics: both are good for SA, and both are bad for wisdom of crowds MAPE. This is exactly the opposite of received wisdom on wisdom of crowds, highlighting the value of replacing anecdotal evidence by data analysis. The method of aggregation is perhaps the most important factor in wisdom of crowds forecasting. Of no surprise to decision theorists, aggregating uncertainty distributions is superior to aggregating point forecasts. Probabilistic crowds are wiser than point forecast crowds. Even if one is interested only in expert-based point predictions, it is preferable to have experts quantify their uncertainty and to combine their distributions with equal weights. Better still is to measure performance as probability assessors and form weighted combinations

Figure 10. (Color online) SA of the Expert in Each Panel with the Best SA Score vs. the MAPE of This Expert

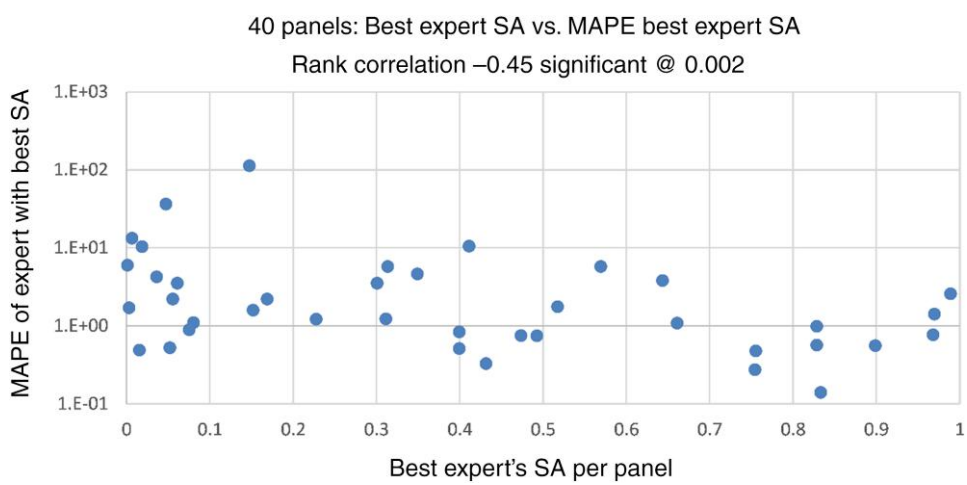


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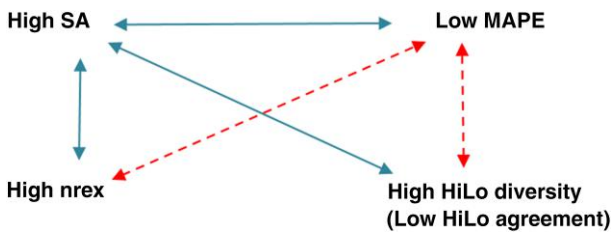
Figure 11. (Color online) Dependence Diagram for Performance Metrics and Number of Experts (nrex) and HiLo Diversity (Low HiLo Agreement)



Note. Solid arrows indicate a "helpful" relation, and dashed arrows indicate a detrimental relation.

of their distributions (Cooke et al. 2021). Measuring performance of uncertainty assessors on calibration variables from the experts' fields not only affords the possibility of performance-based combinations of uncertainty but also enables training in uncertainty quantification. Given all the above, point forecasts are perilous and merit a cautionary advisory. The decision theoretic perspective of extracting point forecasts based on quantification of uncertainty merits wider application.

Acknowledgments

A preliminary version of this paper was presented at the 2025 European Safety and Reliability Conference (Cooke 2025).

Appendix A.

This appendix tabulates the important data for this study, proves the relation between the expert MAPE and WOC MAPE using Jensen's inequality, and defines net agreement and illustrates its calculation. SEJ applications are available online (https://www.cooke-aspinall.net/).

A.1. Application of Jensen's Inequality

Jensen's inequality states that a convex function of an average of arguments is less or equal to the average of the convex function applied to each argument. Absolute value is a convex function.

For a single variable with realization r and n expert forecasts,

$$\left|\left[\left(\sum_{i=1..n} f_i/n\right) - r\right]\Big/ r\right|$$
$$= (1/|r|)\left|(1/n)\left[\left(\sum_{i=1..n} f_i\right) - nr\right]\right|$$
$$= (1/|r|)\left|(1/n)\sum_{i=1..n}(f_i - r)\right|$$
$$\leq (1/(|r|n))\sum_{i=1..n}|(f_i - r)| = (1/n)\sum_{i=1..n}|(f_i - r)/r|.$$

A.2. Agreement−Disagreement for x_1

Matrix $M = M(ex \times vb)$ of $\{-1, 1\}$, nx = number of experts, and nv = number of variables. Net agreement for expert x_1 = NA(1) = #agreements − #disagreements for x_1: $\underline{x}_1 := x_1 \cdot 1 = \sum_{i=1,\ldots,nv} x_{1,i}$. #agreements − #disagreements for $\bar{x}_1$ = NA(1) $= \sum_{i>1} x_1 \cdot x_i$.

$$C_1 := \sum_{i=1\ldots nx} \mathrm{Cov}(x_1, x_i).$$

Lemma A.1. $\mathrm{NA}(1) = nv\, C_1 + (x_1/nv)(\sum_{i\geq 1} x_i) - nv$.

Proof. $C_1 = \mathrm{Cov}(x_1, \sum_{i=1\ldots nx} x_i) = \mathrm{Cov}(x_1, \sum_{i>1} x_i) + \mathrm{VAR}(x_1) = \sum_{i>1} x_1 \cdot x_i/nv - E(x_1)E(\sum_{i>1} x_i) + \mathrm{VAR}(x_1)$. $E(x_1{}^2) = 1$.

$$\sum_{i>1} x_1 \cdot x_i = nv\, C_1 + nv\, E(x_1)E\left(\sum_{i>1} x_i\right) - nv[E(x_1{}^2) - (E(x_1))^2]$$
$$= nv\, C_1 + nv\, E(x_1)\sum_{i\geq 1} E(x_i) - nv$$
$$= nv\, C_1 + (x_1)\left(\sum_{i>1} x_i\right)/nv - nv. \quad \square$$

Table A.1. Studies and Important Covariates

Study	Mean SA	AveCorVbl	AveCorExp	Tail net agr.	HiLo net agr.	nrexp	nrvb	Expert MAPE	WOC MAPE
biol_agents P	5.70E-02	0.00	0.43	0.14	0.29	12	12	5.34	5.01
Brexit_Food P	1.02E-02	0.13	0.08	0.11	0.11	10	10	1.47	0.83
burkina faso PH	4.05E-03	0.05	0.33	0.22	0.24	12	10	310.07	309.65
CARMA PH	2.60E-01	0.15	0.41	0.06	0.24	12	10	248.88	248.26
CDC_ROI PH	1.08E-01	0.09	0.35	0.06	0.23	20	10	3.43	3.05
CDCall PH	1.43E-01	0.03	0.55	0.23	0.38	48	14	6.89	6.71
CO2 E	1.39E-01	−0.07	0.65	0.17	0.40	10	11	0.90	0.82
cotopaxi V	2.30E-01	0.00	0.23	0.15	0.20	20	14	1.66	1.36
DAMS E	3.79E-03	−0.04	0.35	0.20	0.25	11	11	6.56	6.25
Dykering E	4.85E-02	0.15	0.89	0.37	0.80	17	47	0.50	0.48
eBBP P	2.00E-01	0.08	0.32	0.20	0.17	14	15	0.70	0.52

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Table A.1. (Continued)

Study	Mean SA	AveCorVbl	AveCorExp	Tail net agr.	HiLo net agr.	nrexp	nrvb	Expert MAPE	WOC MAPE
Erie Carps E	2.27E-01	0.06	0.49	0.30	0.16	10	15	1.22	1.05
ethiopia PH	8.14E-04	0.01	0.39	0.23	0.24	11	10	37.88	37.58
Gas_3rd_party E	3.77E-03	0.17	0.51	0.18	0.43	14	17	5.56	5.33
GAS95(corros) E	4.62E-02	0.25	0.51	0.21	0.39	13	11	5.75	5.50
Gerstenberger V	6.35E-02	−0.01	0.39	0.14	0.20	12	13	1.40	1.02
ICE_2012 E	8.53E-02	0.05	0.37	0.08	0.30	10	11	1.03	0.82
ICE_2018 E	7.00E-02	−0.03	0.55	0.12	0.41	20	16	1.54	1.04
INFOSEC I	5.31E-02	0.05	0.52	0.13	0.35	13	10	12.24	11.82
Kulumbo V	2.06E-02	0.02	0.42	0.17	0.25	13	19	3.67	3.31
Leontaris E	5.33E-02	0.30	0.60	0.08	0.37	11	14	3.32	3.06
liander E	3.18E-04	−0.02	0.28	0.20	0.35	11	10	1.54	1.35
OPRISKBANK F	1.51E-02	0.06	0.26	0.02	0.14	10	16	332.02	331.55
PHAC_T4 V	9.71E-03	0.09	0.47	0.21	0.33	10	12	56.85	56.57
PILOTS P	3.69E-02	0.02	0.39	0.13	0.31	31	10	0.51	0.29
piton V	4.45E-02	−0.02	0.15	0.04	0.18	28	15	1.04	0.74
Political_Violence P	6.67E-05	0.34	0.44	0.27	0.30	15	21	7.74	7.35
RIVM2023 PH	8.22E-02	0.11	0.52	0.16	0.34	42	15	183.60	183.26
SETE_CIDADES V	1.78E-02	0.04	0.25	0.19	0.16	19	10	877,202.92	877,202.18
Sheep PH	5.64E-02	−0.03	0.40	0.06	0.26	14	15	3.39	3.09
SPEED E	1.83E-02	0.00	0.42	0.10	0.37	14	16	2.14	1.81
Stromboli V	3.95E-02	−0.02	0.27	0.08	0.22	21	16	3.77	3.47
Tadini_Clermont V	2.15E-01	0.02	0.23	0.18	0.23	12	13	2.13	1.94
TdC V	1.03E-01	0.02	0.43	0.00	0.29	18	17	216.32	215.98
TEIDEMAY_05 V	8.32E-03	0.02	0.13	0.06	0.19	17	10	1.49	0.79
Topaz V	3.08E-02	0.02	0.32	0.12	0.30	21	16	49,778.26	49,777.81
TUDDISPR E	4.02E-03	0.22	0.67	0.10	0.40	11	36	0.57	0.45
usgs E	3.15E-03	0.00	0.34	0.10	0.28	32	18	15.55	15.14
VESUVIO V	1.20E-02	−0.01	0.23	−0.01	0.17	14	10	62.36	61.80
Volcano_risk V	3.91E-02	0.00	0.22	0.05	0.19	45	10	316.14	315.45
Average	6.41E-02	0.06	0.39	0.14	0.29	17.45	14.65	23,221.21	23,220.86

Notes. Fields are distinguished as policy (P), public health (PH), engineering/science (E), finance (F), information security (I), and volcanology (V). Expert MAPE is the average of experts' MAPEs; each expert's MAPE is the average absolute percentage error over all variables. WOC MAPE is the average MAPE of the average of experts' medians for each variable. WOC MAPE is less or equal to expert MAPE by Jensen's inequality. Net agreement (agr.) is per expert variable. AveCorVbl (exp) is average correlation between variables (experts).

Remark A.1. The net agreement for expert 1 per variable is $\mathrm{NA}(1)/nv$. The total net agreement for matrix M per (expert, variable) is $\mathrm{NA}(M_{nxnv}) = [\sum_{i=1\ldots nx} \mathrm{NA}(i)]/(nx \times nv)$.

Remark A.2. Setting $q = 1 - p$, the expected net agreement of independent $\{1, -1\}$ variables with $P(1) = p$ is $p^2 + q^2 - 2pq = (p - q)^2 = (2p - 1)^2$. With $(p + q)^2 = 1$, the variance of net agreement is $\mathrm{VAR} = 1 - (2p - 1)^4 = \cdots = 2pq(1 - 4pq)$.

The null hypothesis is that for all cells in M, the probability of one is constant in the set (0, 1) and independent for each cell. Hence, M consists of $nx \times nv$ Bernoulli variables with probability p of "success." For expert 1, the agreement or disagreement with each other expert for each variable constitutes $(nx - 1) \times nv$ independent $\{-1, 1\}$ variables, each with mean $(2p - 1)^2$ and variance $2pq(1 - 4pq)$. The sum of such variables is approximately normal with mean $nv \times (nx - 1) \times (2p - 1)^2$ and standard deviation $[nv \times (nx - 1) \times 2pq \times (1 - 4pq)]^{1/2}$.

A.3. Example Calculation

Table A.2 (top) shows the HiLo matrix for CO2em (Rennert et al. 2022) and the corresponding expert-wise covariance matrix (bottom). The calculation of net agreement for each expert is illustrated.

Table A.3 shows the net agreement for each of the 10 experts in CO2em. For HiLo, two experts' net agreement falls within the 95% central range of the distribution under the null hypothesis; for tail net agreement, only one expert's does.

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Table A.2. HiLo and Covariance Matrices for CO2em

HiLo	vbl1	vbl2	vbl3	vbl4	vbl5	vbl6	vbl7	vbl8	vbl9	vbl10	vbl11	Sum
exp1	−1	1	−1	1	−1	−1	1	1	−1	1	−1	−1
exp2	1	−1	−1	−1	−1	1	1	1	−1	1	−1	−1
exp3	1	−1	−1	−1	−1	−1	1	1	−1	1	−1	−3
exp4	1	−1	−1	1	−1	1	1	1	−1	1	−1	1
exp5	−1	−1	−1	−1	1	−1	1	1	−1	1	1	−1
exp6	1	−1	−1	−1	1	−1	−1	1	−1	1	1	−1
exp7	1	−1	−1	1	−1	−1	−1	1	−1	1	−1	−3
exp8	−1	−1	−1	−1	−1	1	1	1	−1	1	−1	−3
exp9	1	−1	−1	1	−1	−1	1	1	−1	1	1	1
exp10	1	−1	−1	−1	1	−1	−1	1	−1	1	−1	−3
												−14

HiLo covariance	exp1	exp2	exp3	exp4	exp5	exp6	exp7	exp8	exp9	exp10
exp1	0.99	0.26	0.43	0.46	0.26	−0.10	0.43	0.43	0.46	0.07
exp2	0.26	0.99	0.79	0.83	0.26	0.26	0.43	0.79	0.46	0.43
exp3	0.43	0.79	0.93	0.66	0.43	0.43	0.56	0.56	0.66	0.56
exp4	0.46	0.83	0.66	0.99	0.10	0.10	0.66	0.66	0.63	0.30
exp5	0.26	0.26	0.43	0.10	0.99	0.63	0.07	0.43	0.46	0.43
exp6	−0.10	0.26	0.43	0.10	0.63	0.99	0.43	0.07	0.46	0.79
exp7	0.43	0.43	0.56	0.66	0.07	0.43	0.93	0.20	0.66	0.56
exp8	0.43	0.79	0.56	0.66	0.43	0.07	0.20	0.93	0.30	0.20
exp9	0.46	0.46	0.66	0.63	0.46	0.46	0.66	0.30	0.99	0.30
exp10	0.07	0.43	0.56	0.30	0.43	0.79	0.56	0.20	0.30	0.93
sumCov	3.70	5.52	6.02	5.39	4.07	4.07	4.93	4.56	5.39	4.56
Net agr.	31	51	59	47	35	35	47	43	47	43

Notes. The net agreement per expert is computed as *nrvb*sumCov + expsum HiLo*sumHiLoMatrix/nrvb − nrvb. Here Sum[matrix] denotes the sum of all enties in the matrix.* For expert 1 (exp1), this is 11 × 3.702479339 + (–1) × (–14)/11 – 11 = 31. vbl, variable. sumCov denotes the sum of terms in a covariance matrix.

Table A.3. Net Agreement for Each of the 10 Experts in CO_2m (11 Variables)

Expert	HiLo net agreement	*ExpSumNA HiLo*	Normal CDF	Tail net agreement	*ExpSumNA Tail*	Normal CDF
1	0.28	−1	4.82E-04	0.34	3	9.99E-01
2	0.46	−1	4.82E-04	0.11	−5	1.28E-26
3	0.54	−3	2.67E-09	0.00	−7	3.86E-45
4	0.43	1	2.22E-01	−0.07	−3	3.54E-13
5	0.32	−1	4.82E-04	0.01	−1	9.13E-05
6	0.32	−1	4.82E-04	0.34	5	1.00E+00
7	0.43	−3	2.67E-09	0.16	1	3.80E-01
8	0.39	−3	2.67E-09	0.31	5	1.00E+00
9	0.43	1	2.22E-01	0.34	5	1.00E+00
10	0.39	−3	2.67E-09	0.16	9	1.00E+00

Note. ExpSumNA denotes the sum per expert of net agreement.

Appendix B. Primer: Fat-Tailed Distributions[6]

In 1898, Italian engineer/economist Vilfredo Pareto (1898) introduced a distribution to describe wealth in society, noting that 80% of the wealth was typically owned by 20% of the population. For $\alpha > 0$, its complementary cumulative distribution function C, also known as its survivor function (one minus its cumulative distribution function F), is given by

$$P(X > x) = 1 - F(x) = C(x) = x^{-\alpha};\ x > 1. \tag{B.1}$$

The exponent α is called the *tail index*. The density function is $f(x) = \alpha x^{-(\alpha+1)}$. Its κth moment is

$$\int_{x=1\ldots\infty} x^k \alpha x^{-(\alpha+1)} dx, \tag{B.2}$$

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which is infinite for $k \geq \alpha$ and finite for $\kappa < \alpha$. For $\alpha = 2$, the variance is infinite, and the mean is finite. This means the average of n independent copies, $(1/n)\sum_{i=1\ldots n} X_i$, has finite mean but infinite variance. As $n \to \infty$, the average does not converge to the mean but converges to a random variable that also has infinite variance. Embrechts et al. (2003) identify several classes of fat-tailed distribution, the most important of which are discussed briefly below. Kurtosis, the fourth standardized moment, is an older notion of tailedness. It has passed out of favor but survives in the notion of "leptokurtic" indicating "excess peakedness and heavy tails" (Embrechts et al. 2003, p. 407).

A *regularly varying distribution* is one whose survivor function is asymptotically Pareto. More precisely,

$$\lim_{t\to\infty} C(tx)/C(x) = t^{-\alpha}, \quad \alpha > 0,\ x > 1. \tag{B.3}$$

We define R_α as the class of regularly varying distributions with tail index α. For $\kappa > \alpha$, the κth moments are infinite; for $\kappa < \alpha$, they are finite. The case $\kappa = \alpha$ is indeterminate; depending on the distribution, it can be either finite or infinite. If $\alpha = 0$, the function is *slowly varying* (asymptotically constant); for $\alpha = \infty$, the distribution is *rapidly varying*, and all moments are finite.

The following regularly varying distributions are important in extreme value theory.

The first is the *Fréchet distribution*:

$$F(x) = exp(-x^{-\alpha}): \quad \text{for } x > 0,\ \alpha > 0. \tag{B.4}$$

Under appropriate scaling, the maxima of a regularly varying distribution converge to a Fréchet distribution as $x \to \infty$. Note that for large x, $exp(-x^{-\alpha})$ is approximately equal to $1 - x^{-\alpha}$; hence the Fréchet distribution converges to the Pareto as $x \to \infty$. Introducing scale and location parameters, (B.4) becomes the *generalized extreme value distribution* (*GEV*):

$$\begin{aligned} &F(x;\mu,\sigma,\varepsilon) \\ &\quad = exp\{-[1+\varepsilon(x-\mu)/\sigma]^{-1/\varepsilon}\}: \quad \text{for } 1+\varepsilon(x-\mu)/\sigma > 0, \varepsilon \neq 0, \\ &\quad = exp(-exp(-x));\ \text{for } \varepsilon = 0. \end{aligned} \tag{B.5}$$

The *generalized Pareto distribution* (*GPD*) also features scale and location parameters:

$$\begin{aligned} F(x;\mu,\sigma,\varepsilon) &= 1 - [1+\varepsilon(x-\mu)/\sigma]^{-1/\varepsilon}: \quad \text{for } \varepsilon \neq 0, x > \mu \\ &\qquad \text{when } \varepsilon \geq 0,\ x \leq \mu - \sigma/\varepsilon \text{ when } \varepsilon < 0 \\ &= 1 - exp(-(x-\mu)/\sigma): \text{for } \varepsilon = 0. \end{aligned} \tag{B.6}$$

The GPD is used in estimating the distribution of peaks over threshold (Embrechts et al. 2003). If $\varepsilon > 0$, the GEV and GPD are regularly varying with tail index $1/\varepsilon$ and differ in the way they approach Pareto distributions as $x \to \infty$. The term ε is sometimes called the *extreme value index*. Note that the parameters in the above distributions describe limiting distributions as $x \to \infty$. Any procedure for estimating these parameters must invoke heuristics to determine a threshold above which values of x are used to estimate the parameters.

An important property of regularly varying distributions is the following. Let F^{n^*} and C^{n^*} denote the cumulative distribution function and survivor function, respectively, of the n-fold convolution of independent and identically distributed (iid) copies of a random variable $X \in R_\alpha$. For regularly varying distributions, $lim_{x\to\infty} C^{n^*}(x)/C(x) = n$, or, written out,

$$\lim_{x\to\infty} \frac{Prob(X_1 + \cdots + X_n > x)}{Prob(X > x)} = n. \tag{B.7}$$

Noting that $Prob(Max\{X_1, \ldots, X_n\} > x) = 1 - F^n(x) = (1 - F(x)) \sum_{k=0\ldots(n-1)} F(x)^k \to n(1-F(x))$ as $x \to \infty$, we have (Embrechts et al. 2003)

$$\lim_{x\to\infty} Prob(X_1 + \cdots + X_n > x) = Prob(Max\{X_1, \ldots X_n\} > x). \tag{B.8}$$

This says that the sum behaves asymptotically like the max. This holds for all regularly varying distributions regardless of the tail index. The lower the tail index, the fatter the tail, and the sooner this behavior becomes manifest as $x \to \infty$. This motivates the obesity index, introduced below, applying equally to distribution functions and finite iid samples.

The *subexponential* distributions emerge in modeling insurance claims and are always nonnegative.

Random variable $X > 0$ with survivor function $C(x)$ is *subexponential* if $lim_{x\to\infty} C^{n^*}(x)/C(x) = n$. This converts property (B.7) to a definition for positive variables. In fact, if (B.7) holds for $n = 2$, then it holds for all n. All subexponential variables have property (B.8) that the sum of independent copies behaves asymptotically like the max, and all positive regularly varying distributions aresubexponential. However, there are rapidly varying distributions (all moments are finite) that are also subexponential, including the log-normal and Weibull for appropriate parameter values. These distributions will exhibit fat-tail behavior for finite samples but will eventually converge all moments, while preserving the *sum ~ max* relation.

The subexponential class suggests an obesity index that can be computed for any density function and also estimated by bootstrapping for any finite sample distribution (Cooke et al. 2014, see also Xu et al. 2025).

The idea is as follows; denote by $(X(1), X(2), X(3), X(4))$ four independent samples of positive variable X, arranged so that $X(1) \leq X(2) \leq X(3) \leq X(4)$. If the max $X(4)$ is nearly equal to $X(4) + X(3) + X(2) + X(1)$, then $X(4) \gg X(3) + X(2) + X(1) > X(3) + X(2) - X(1)$. In that case, $X(4) + X(1) > X(3) + X(2)$. We therefore define the obesity index as

$$\begin{aligned} Obx(X) &= Prob(X_{(4)} - X_{(3)} > X_{(2)} - X_{(1)}) \\ &= Prob(X_{(4)} + X_{(1)} > X_{(2)} + X_{(3)}). \end{aligned} \tag{B.9}$$

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Table B.1. Obesity Index for Pareto Distribution with $\alpha = 1$, 2, 3, 4

α	Exact value	Approximate value
1	$\pi^2 - 9$	0.8696
2	$593 - 69\pi^2$	0.8237
3	$-124{,}353/5 + 2{,}520\pi^2$	0.8031
4	$19{,}150{,}997/21 - 92{,}400\pi^2$	0.7912

For any random variable X (not necessarily positive) with survivor function C and density f, we have

$$Obx(X) = 24 \int_{x1=-\infty\ldots\infty} \int_{x1\ldots\infty} \int_{x2\ldots\infty} C(x_2 + x_3 - x_1) f(x_1) f(x_2) f(x_3) dx_3 dx_2 dx_1, \quad \text{(B.10)}$$

allowing Obx to be numerically computed for X. The innermost integral is the probability that the random variable X is larger than $x_3 + x_2 - x_1$. Cooke et al. (2014) prove inter alia the following: For any positive random variable X and any g: $\mathbb{R}_+ \to \mathbb{R}_+$ strictly convex increasing, $Obx(X) < Obx(g(X))$. For Pareto(α) variable X, $lim_{\alpha \to 0}\ Obx(X) = 1$. The Obx for the uniform distribution on a positive interval is 0.5; for the absolute value of a normal distribution, it is 0.66; for the exponential distribution, it is 0.75. For any symmetric distribution, it is 0.5 (though it is intended for positive variables). Exact and approximate values of Obx for a few Pareto distributions are given in Table B.1.

Figure B.1 shows obesity as a function of distribution parameters for Pareto, Weibull, and log-normal distributions. For the latter two, the variance goes to infinity as the obesity goes to one. From the obesity viewpoint, the difference between having infinite versus finite variance disappears as the variance gets "close to infinity."

Figure B.1. (Color online) Obesity Indices for (a) Pareto, (b) Weibull, and (c) Log-Normal as Functions of the Distributions' Parameters (from Cooke et al. 2014, figure 5.7)

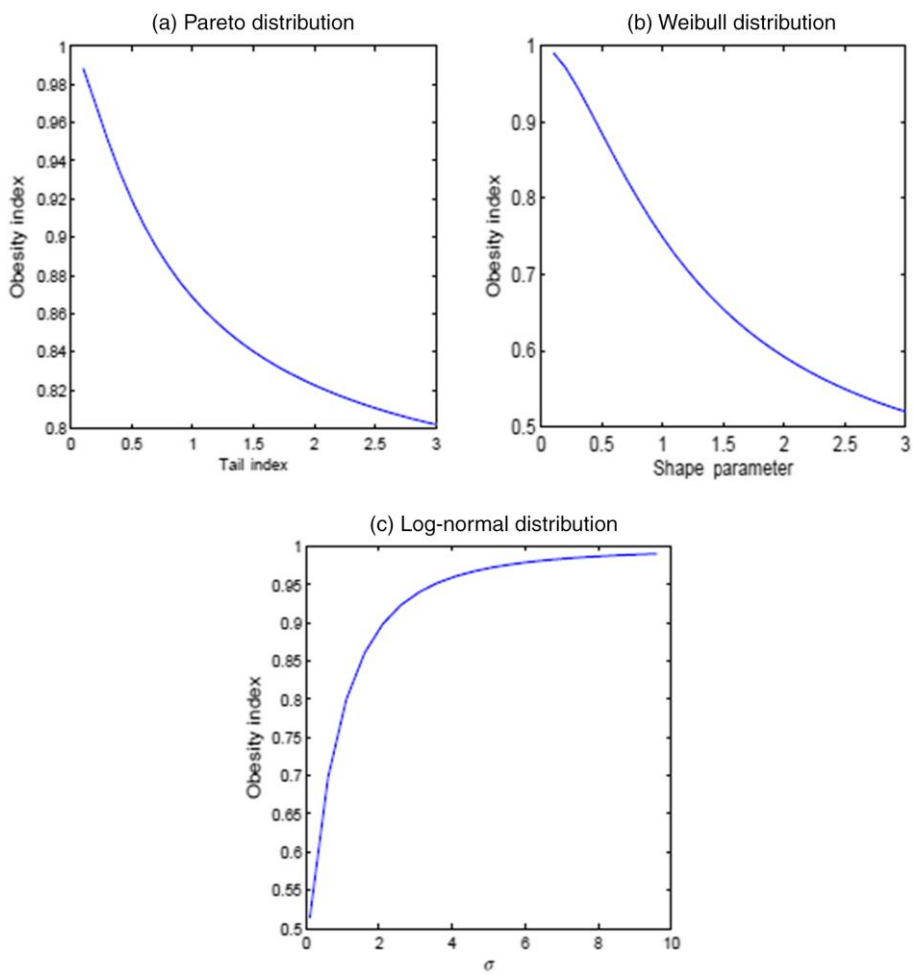

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Table B.2. Obesity Index for Five Sets of MAPEs with 90% Confidence Bounds Based on 10,000 Sampled 4-Tuples

Obx 586 MAPEs nr exp ≥ 10
a. 0.895 (0.890, 0. 900)
Obx 586 Sym MAPEs
b. 0.881 (0.876, 0.886)
Obx 842 MAPEs nr exp < 10
c. 0.880 (0.875, 0.885)
Obx 432 MAPEs nr exp $= 10, \ldots, 19$
d. 0.8901 (0.8849, 0.8952)
Obx 154 MAPEs nr exp ≥ 20
e. 0.8983 (0.8933, 0.9032)

For a finite iid sample of X, we can estimate $Obx(X)$ by bootstrapping. The number of 4–tuples for modestly sized samples is large, with 586 samples there are 4.86E9 4-tuples. The confidence bounds around such estimates become very narrow as the number of sampled 4-tuples increases. Table B.2 shows the obesity index with 90% confidence bounds based on sampling 100,004-tuples for

a. 586 WOC MAPEs from the 40 panels with at least 10 experts and 10 variables,

b. Sym MAPEs for the same 586 samples,

c. 842 WOC MAPEs from all panels in the complementary set of 67 $(107 - 40)$ studies,

d. 154 WOC MAPEs from the subset of 586 with at least 20 experts, and

e. 432 WOC MAPEs from the subset of 586 with 10 up to 19 experts.

The obesity of 586 MAPEs corresponds to that of a Pareto($\alpha = 0.7$). Note that all five obesity values are higher than that of Pareto($\alpha = 1$). The Sym MAPE values are significantly less obese than the MAPE values, though still quite obese, and the same holds for the 842 MAPEs: the latter are significantly less obese than 586 MAPEs, but still quite obese. The subsets of the 586 MAPEs with fewer than 20 and at least 20 experts are not statistically distinguishable on these 100,004-tuples as the 90% confidence intervals overlap. To distinguish these values, we need to sample 50,000 k-tuples (not shown).

Endnotes

[1] Cooke and Goossens (2008) references 45 pre-2006 studies, and Cooke et al. (2021) references 49 post-2006 studies. The remaining are published but not bundled. SEJ applications are available online (https://www.cooke-aspinall.net/).

[2] Of these, 60 have fewer than 10 experts.

[3] Independence in this context is not an assumption about an expert's subjective distribution but a desideratum of the scoring (Cooke 1991).

[4] This is a nontechnical introduction to fat-tailed distributions. Many textbooks give a full mathematical treatment. Cooke et al. (2014) is directed to numerate nonspecialists. Appendix B gives more technical background.

[5] This study subsetted the post-2006 studies, which were of more uniform design than many of the pre-2006 studies.

[6] This primer is based on Cooke et al. (2014) and Embrechts et al. (2003).

References

Aspinall WP (2006) Structured elicitation of expert judgment for probabilistic hazard and risk assessment in volcanic eruptions. *Statistics in Volcanology* (Geological Society, London), 15–30.

Aspinall WP (2010) A route to more tractable expert advice. *Nature* 463(7279):294–295.

Aspinall WP, Loughlin SC, Michael FV, Miller AD, Norton GE, Rowley KC, Sparks RSJ, Young SR (2002) The Montserrat Volcano Observatory: Its evolution, organization, role and activities. Druitt TH, Kokelaar BP, eds. *The Eruption of Soufrière Hills Volcano, Montserrat, from 1995 to 1999* (Geological Society, London).

Bruine de Bruin W, van der Klaauw W, Topa G (2011) Expectations of inflation: The biasing effect of thoughts about specific prices. *J. Econom. Psych.* 32(5):834–845.

Burgman M, Carr A, Godden L, Gregory R, McBride M, Flander L, Maguire L (2011) Redefining expertise and improving ecological judgment. *Conservation Lett.* 4(2):81–87.

Colson AR, Cooke RM (2017) Cross validation for the classical model of structured expert judgment. *Reliability Engrg. System Safety* 163:109–120.

Colson AR, Cooke RM (2018) Expert elicitation: Using the classical model to validate experts' judgments. *Rev. Environ. Econom. Policy* 12(1):113–132.

Cooke RM (1991) *Experts in Uncertainty: Opinion and Subjective Probability in Science* (Oxford University Press, Oxford, UK).

Cooke RM (2022) Averaging quantiles, variance shrinkage and overconfidence. *Futures Foresight Sci.* 5(1):e139.

Cooke RM (2025) Wisdom or madness; expert data on wisdom of crowds. Abrahamsen EB, Aven T, Bouder F, Flage R, Ylönen M, eds. *Proc. 35th Eur. Safety Reliability 33th Soc. Risk Anal. Eur. Conf.* (Research Publishing, Singapore).

Cooke RM, Goossens LHJ (2008) TU Delft expert judgment data base. *Reliability Engrg. System Safety* 93(5):657–674.

Cooke RM, Marti D, Mazzuchi TA (2021) Expert forecasting with and without uncertainty quantification and weighting: What do the data say? *Internat. J. Forecasting* 37(1):378–387.

Cooke RM, Nieboer D, Misiewicz J (2014) *Fat-Tailed Distributions: Data, Diagnostics and Dependence* (Wiley, Hoboken, NJ).

Eggstaff JW, Mazzuchi TA, Sarkani S (2014) The effect of the number of seed variables on the performance of Cooke's classical model. *Reliability Engrg. System Safety* 121:72–82.

Embrechts P, Kluppelberg C, Mikosch T (2003) *Modelling Extremal Events for Insurance and Finance* (Springer–Verlag, Berlin).

Flores BE (1986) A pragmatic view of accuracy measurement in forecasting. *Omega* 14(2):93–98.

Galton F (1907) Vox populi. *Nature* 75:450–451.

Gneiting T, Balabdaoui F, Raftery AE (2007) Probabilistic forecasts, calibration and sharpness. *J. R. Statist. Soc. B* 69(Part 2):243–368.

Gupta N, Chavan SR (2021) Characterizing the tail behaviour of daily precipitation probability distributions over India using the obesity index. *Internat. J. Climatology* 42(4):2543–2565.

Hald T, Aspinall W, Devleesschauwer B, Cooke RM, Corrigan T, Havelaar AH, Gibb H, et al. (2016) World Health Organization estimates of the relative contributions of food to the burden of

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disease due to selected foodborne hazards: A structured expert elicitation. *PLoS One* 11(1):e0145839.

Hanea AM, McBride MF, Burgman MA, Wintle BC (2018) The value of performance weights and discussion in aggregated expert judgments. *Risk Anal.* 38(9):1781–1794.

Kendall MG, Kendall SFH, Babington Smith B (1939) The distribution of spearman's coefficient of rank correlation in a universe in which all rankings occur an equal number of times. *Biometrika* 30(3/4):251–273.

Lichtendahl KC Jr, Grushka-Cockayne Y, Winkler RL (2013) Is it better to average probabilities or quantiles? *Management Sci.* 59(7):1594–1611.

Mackay C (1841) *Memoirs of Extraordinary Popular Delusions and the Madness of Crowds*, vol. I (Richard Bentley, London).

Morley SK, Brito TV, Welling DT (2018) Measures of model performance based on the log accuracy ratio. *Space Weather* 16(1):69–88.

Murray D (2019) *The Madness of Crowds: Gender, Race and Identity* (Bloomsbury, London).

Niu X, Harvey N (2022) Point, interval, and density forecasts: Differences in bias, judgment noise, and overall accuracy. *Futures Foresight Sci.* 4(3–4):e124.

Oreskes N (2019) *Why Trust Science?* (Princeton University Press, Princeton, NJ).

Pareto V (1898) Cours d'economie politique. *J. Political Econom.* 6:549–552.

Planck MK (1950) *Scientific Autobiography and Other Papers* (Philosophical Library, New York).

Rennert K, Errickson F, Prest BC, Rennels L, Newell RG, Pizer W, Kingdon C, et al. (2022) Comprehensive evidence implies a higher social cost of CO_2. *Nature* 610(7933):687–692.

Surowieki J (2005) *The Wisdom of Crowds*, reprint ed. (Anchor, New York).

Taleb N (2007) *The Black Swan: The Impact of the Highly Improbable* (Random House, New York).

Thomas RP, Lawrence A (2018) Assessment of expert performance compared across professional domains. *J. Appl. Res. Memory Cognition* 7(2):167–176.

Weiss DJ, Shanteau J (2004) The vice of consensus and the virtue of consistency. Smith K, Shanteau J, Johnson P, eds. *Psychological Investigations of Competence in Decision Making* (Cambridge University Press, Cambridge, UK), 226–240.

Wietzke LM, Merz B, Gerlitz L, Kreibich H, Guse B, Castellarin A (2020) Comparative analysis of scalar upper tail indicators. *Hydrological Sci. J.* 65(10):625–1639.

Winkler RL, Grushka-Cockayne Y, Lichtendahl KC Jr, Jose R, Victor R (2019) Probability forecasts and their combination: A research perspective. *Decision Anal.* 16(4):239–260.

Xu TL, Rodica Ioana D, Florin S (2025) Identifying fat-tailed distributions: Methods and insights. *10th Internat. Conf. Math. Inform.* (Centre for the Study of Complexity, Babeş-Bolyai University, Cluj Napoca, Romania).

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